Sungwon Kim

https://sungwon-kim.com | 01) 917-454-8696 | skim434@jhu.edu | https://github.com/blindTissue

EDUCATION

Johns Hopkins University ***Baltimore, MD***

Masters in Computer Science *Graduated: May 2025*

- Graduated with Computer Science Masters at Johns Hopkins University.

Johns Hopkins University ***Baltimore, MD***

BS Computer Science, Applied Mathematics & Statistics *Graduated: May 2024*

- GPA: 3.90/4.0
- Course Assistant for NLP: Self-supervised Models, Natural Language Processing, Introduction to Algorithms, Automata and Computation Theory, Discrete Math.
- Math Tournament Problem Writer: ASDAN, JHMT (Johns Hopkins Math Tournament).

Work Experience

Johns Hopkins University ***Baltimore, MD***

Research Assistant *August 2025 - Current*

- Research Assistant under Professor Daniel Khashabi

DataTechnica ***Bethesda, MD***

Part Time *August 2024 - December 2024*

- Engineered a medical RAG(Retrieval Augmented Generation) pipeline. Implementing various methods for SQL retrieval including chain-of-thought reasoning and self-consistency.
- Helped create datasets and evaluation metrics for medical questions.

National Institutes of Health ***Bethesda, MD***

Research Intern *May 2024 - August 2024*

- Worked on natural language to SQL query pipeline that scales sublinearly to the number of columns
- Achieved 5% increase in accuracy (80% to 85%) compared to vanilla schema-as-context methods.

Center for Language and Speech Processing ***Baltimore, MD***

Research Assistant *May 2023 – May 2024*

- Created chatbots which cite databases for a web based app.
- Developed and tested various retrieval methods including dense, sparse, hybrid retrieval methods, chunking strategies, LLM fine tuning, for robust search using Huggingface, sbert, Apache Solr, nltk, and more.

Johns Hopkins University ***Baltimore, MD***

Research Assistant *May 2023 – September 2023*

- Created neural network-derived brain perfusion map processing pipeline for model-free CTP images.
- Achieved up to 30% better MSE loss compared to a recent paper on the same dataset.

Republic of Korea Army ***Yanggu, Korea***

Information Systems Maintenance/Operator, Squad Leader *September 2020 – March 2022*

- Maintained and operated secure military information systems to ensure reliable communication of sensitive data.
- Appointed Squad Leader in June of 2021.

Nexon Korea ***Pangyo, Korea***

Platform Analyst (Intern) *June 2020 – August 2020*

- Analyzed game data logs using SQL, Hadoop, and Python.
- Built ML model (Random Forest, LSTM, XGBoost) to classify player experience levels, improving accuracy by 15%+ over a Markov Chain baseline.

Publications

Afterglow: Forward Span Propagation for Query-Aware KV Cache Eviction

Submitted to NAACL 2027

- Improves published query-aware KV cache eviction by propagating high attention score with decay to later tokens.
- Currently scores first in kvpress leaderboard for query-aware single pass eviction method.

VFold: Symmetry-Aware Cross-Layer Value Cache Compression

Submitted to ICLR 2027

- Inter-layer cache merging algorithm that exploits cross-layer redundancy.

Sparse Attention with Entropy-Triggered KV Repair

Submitted to ICLR 2027

- Entropy triggered sparse attention cache rebuild for accurate responses.

Recovering Collapsed Language Models in Weight Space Without External Data
Submitted to EACL 2027

- Experimented multi-model feedback loop and confirmed that model collapse appears in this scenario.
- Found that collapse is locally reversible via task vector negation.

What Survives Eviction? Information Recall Locality in Transformer KV Caches
Accepted to TMLR, 2026

- Explored cache locality in information retrieval.

BiomedBench Suite: Benchmarks for Evaluating LLM Performance on Biomedical Reasoning
Accepted to ICML 2026 Workshop

BiomedSQL: Text-to-SQL for Scientific Reasoning on Biomedical Knowledge Bases
Accepted to COLM 2026

- Dataset for evaluating scientific reasoning in text-to-SQL generation over a real-world biomedical knowledge base.

CARDBiomedBench: A Benchmark for Evaluating Large Language Model Performance in Biomedical Research
Lancet Digital Health, 2026

- Large Scale Question and Answer Benchmark for evaluating LLMs in biomedical science.

Challenging the Evaluator: LLM Sycophancy Under User Rebuttal
Accepted to EMNLP 2025 Findings

- Investigated qualities that trigger LLM sycophancy under user rebuttal.

PROJECTS

Logit Lens Llama Advanced *October 2025 - December 2025*

- Created an interactive version of Logit Lens, where users can intervene on residual, attention and FFN stream, on a per-layer, per-head basis.

Embedding-Level Translation for Seamless Communication Across LLM Agents *December 2024*

- Developed an embedding-level translator to enable seamless communication between LLM agents, achieving <0.01 MSE and >0.9 cosine similarity, demonstrating high-fidelity translation across agent representations.

Linear Embedding Space Alignment for Semantic Change Detection *June 2024*

- Explored limitations of the Procrustes alignment method for detecting domain shifts in semantic spaces over time.
- Proposed and evaluated alternative linear alignment techniques using unsupervised anchor discovery, tested on a machine translation benchmark.
- Found that while methods varied in approach, no significant performance difference was observed across alignment strategies.

arXiv Reference Finder (Hophacks - 2nd Place Winner) *September 2023*

- Built a tool to extract and highlight relevant sections from referenced papers on arXiv, given a source paper.
- Utilized BM25 and cross-encoders for semantic similarity; integrated arXiv API and applied regex-based PDF chunking and formatting.

Custom Transformers *May 2023 – June 2023*

- Implemented transformer layers with non-standard architectures.
- Achieved lower classification loss than HuggingFace BERT, despite using fewer parameters.

Misc Game Tools(Toys) *June 2020 – June 2022*

- Designed an Excel-based Markov chain model to calculate expected costs of in-game equipment upgrades.
- Developed a course planner to track and store all Johns Hopkins University course data locally.
- Created a Pokémon battle damage calculator and derived a custom method for type ranking analysis.

SKILLS

Programming: Python, Pytorch, HuggingFace Transformers, scikit-learn, pandas, numpy, C++
Software/Skill set: LaTeX, SQL, Microsoft Office (Excel, PowerPoint, Word, Outlook)